

ZISAN AALAM

Full Stack Software Engineer | Java • Python • React • JavaScript • AWS • Microservices

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 3+ years of professional experience designing, developing and supporting enterprise web applications, REST APIs, microservices and database-driven systems within a US-based healthcare technology company. Experienced across Angular, React, backend development, relational databases, system integrations, AWS, Docker, Kubernetes and CI/CD. Strong background in solution design, database and framework modernization, testing, deployment and production support. Experienced working with cross-functional stakeholders to translate business requirements into reliable technical solutions, with practical use of AI-assisted development tools to improve engineering productivity and software quality.

TECHNICAL SKILLS

Full Stack Development:	Angular, React, JavaScript (ES6+), TypeScript, HTML5, CSS3, Responsive Web Development
Backend & Integration:	C#, .NET, Java, Python, Node.js, REST APIs, Microservices, Enterprise System Integration, Asynchronous & Event-Driven Systems
Databases & Data:	PostgreSQL, MySQL, MongoDB, SQLite, Redis - Relational Database Design, SQL, Data Modelling, Query Design & Performance Optimizations
Cloud & DevOps:	AWS, Kubernetes, Docker, GitHub Actions, Jenkins, Git, CI/CD, Automated Deployment
Software Engineering:	Solution Design, Enterprise Application Development, Automated Testing, Code Reviews, Debugging, Release Management, Production Support, Technical Documentation
Security & Reliability:	Secure Coding, API Security, Authentication & Authorisation, Encryption, Production Support, Debugging, Performance & Reliability
AI-Assisted Development:	Claude, GitHub Copilot, Cursor, ChatGPT, Gemini - coding, debugging, refactoring, test generation and technical analysis
Collaboration & Tools:	Jira, Confluence, Redmine, IntelliJ IDEA, VS Code, pgAdmin, Docker Desktop, Jupyter Notebook

WORK EXPERIENCE

Software Engineer

Feb 2022 – Feb 2025

Cedar Gate Technologies (US-based healthcare technology company)

Kathmandu, Nepal

- ▶ Led the end-to-end design and development of a microservice using React, Node.js, and MongoDB to generate business reports by aggregating data from Elasticsearch and Jira -directly aligning with the requirement to design and deliver bespoke applications.
- ▶ Drove optimisation of database data flows by successfully migrating MySQL from v5 to v8, redesigning slow queries, and improving retrieval speeds - demonstrating SQL proficiency including joins, stored procedures, and query design.
- ▶ Designed and implemented automated workflows and task scheduling scripts using Bash and Python, reducing manual effort and improving operational efficiency across multiple platforms.
- ▶ Contributed to architectural design of automated jobs and participated in release management, testing, and change control processes - with clear documentation and stakeholder consultation for each deployment.
- ▶ Delivered features and resolved defects across microservices built on Node.js, RxJava, Vert.x, and Grails, ensuring seamless integration across Windows and Linux environments.
- ▶ Developed front-end components in React and Angular, collaborating with cross-functional teams to deliver responsive, user-friendly interfaces.
- ▶ Managed and maintained AWS infrastructure including EC2, Lambda, API Gateway, S3, SQS, ECR, and ECS - supporting application deployments and cloud operations.
- ▶ Performed a major Vert.x framework upgrade (v3.3.2 → v4.5.10) across multiple services, managing dependencies and ensuring backward compatibility.
- ▶ Participated in code reviews, test planning, and debugging cycles to uphold code quality and application reliability.
- ▶ Provided high-level internal support for operational issues, coordinating with vendors and ensuring timely resolution.
- ▶ Migrated Redmine issue-tracking integrations to Jira across multiple services, demonstrating adaptability and vendor tool management.

Software Engineer Intern

Dec 2021 – Jan 2022

Leapfrog Technology

Kathmandu, Nepal

- Built pixel-perfect responsive web pages using HTML, CSS, and vanilla JavaScript; created lo-fi prototypes and designs in Figma for the TuTangle platform.
- Completed a Postman-clone capstone project using HTML, CSS, JavaScript, and the Canvas API demonstrating full front-end development capability.

UNIVERSITY PROJECTS

ICT Master Capstone Project - SA Power Networks | *Adelaide University (2026 - In Progress)*

Collaborating with SA Power Networks on machine-learning-based data imputation for missing asset attributes used in probability-of-failure modelling, involving dataset analysis, comparative modelling and industry stakeholder collaboration.

RAG System for Cybersecurity Guidelines | *Advanced Topics in AI & Machine Learning, University of Adelaide (2026)* Built a domain-specific Retrieval-Augmented Generation system to generate actionable phishing defence guidelines for cybersecurity practitioners. Designed the full pipeline including document chunking, embedding with all-mpnet-base-v2, vector storage in ChromaDB, and guideline generation using Mistral-7B-Instruct via Ollama. Evaluated system trustworthiness using the RAGAS framework across context relevance, answer relevance, and faithfulness metrics. Solo project.

Secure Overlay Chat Protocol (SOCP) - Distributed Messaging System | *Secure Programming, Adelaide University (2025)*

Developed a secure, distributed multi-server chat application in Python implementing end-to-end encrypted messaging, cross server user routing, and file transfer with SHA-256 hash verification. Built using WebSocket, FastAPI, and PyCryptodome (RSA4096), featuring deadlock-safe message routing across a 3-server mesh network, heartbeat-based connection management, server failure recovery, and cryptographic signature verification on all inter-server communication. Team project.

Flappy Bird Deep Q-Network (DQN) Agent | *Artificial Intelligence, Adelaide University (2025)*

Designed and trained a Deep Q-Network reinforcement learning agent in Python to autonomously play Flappy Bird, successfully completing all 7 difficulty levels including a secret level beyond the assessed scope. Implemented experience replay, epsilon greedy exploration, dual-network architecture (Q and target network), and a custom 8-feature state representation with reward shaping. Built using PyTorch, NumPy, and Gymnasium. Solo project.

Petri Net Simulation - Adelaide Metro Train Scheduling | *Event-Driven Computing, Adelaide University (2025)*

Designed and implemented a Petri Net simulation of the Adelaide Metro network in Java, modelling concurrent train scheduling with collision prevention, deadlock detection, directional flow control, and correct arrival/exit handling across shared track sections. Solo project.

EDUCATION

Master of Computer Science | GPA: 6.417 / 7.0

2025 – Current

Adelaide University

Adelaide, SA, Australia

Current: ICT Master Capstone Project, Generative AI, Computer Networking. Relevant study: Event-Driven Computing (HD, 86%), Distributed Systems (D, 82%), Secure Software Engineering (D, 81%), Secure Programming (HD, 86%), Specialized Programming (HD, 85%).

Bachelor of Computer Science and Engineering | CGPA: 8.47 / 10

2017 – 2021

Vellore Institute of Technology

Vellore, India

Scholarship: COMPEX-NEPAL

CERTIFICATIONS

- Introduction to Data Science in Python - University of Michigan (Jan 2022)
- Server-side Development with Node.js, Express, and MongoDB - HKUST (Oct 2019)
- Single Page Web Applications with AngularJS - Johns Hopkins University (Mar 2019)
- Java for Android - Vanderbilt University (Apr 2020)
- Digital Footprint - University of Edinburgh (Jun 2020)

References available upon request